The Ramanujan Protocol:
A Self-Correcting Pipeline for Data-Driven Conjecture via Symbolic Regression and Formal Verification

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Abstract

We present the *Ramanujan Protocol*, a closed-loop pipeline for automated mathematical conjecture that **observes** numerical data, **discovers** formulas via genetic programming, **verifies** them through Z3 SMT solving, and **self-corrects** when findings are structurally trivial or statistically unsupported.

The system's principal contribution is *epistemic self-correction*: it autonomously caught a GCT obstruction "discovery" that was a structural tautology, and a "universal power-law exponent" for *abc* triple quality that was an artifact of model choice (exponential decay $R^2 = 0.93$ vs. power law $R^2 = 0.69$, AIC difference 596). Both were retracted before publication.

As a test of cross-domain pattern matching, the system recovered all 7 known modularity correspondences between elliptic curves and modular forms from raw coefficient data (95/95 prime matches, 0/3 decoy false positives), confirmed the Sato–Tate $\sin^2\theta$ distribution from 302 primes, and computed BSD critical values matching LMFDB to 4 significant figures. It formally proved $R(3,3) = 6$ and $R(3,4) = 9$ via SAT encoding, and verified the Montgomery pair correlation against GUE ($\chi^2 = 9.28$, $df = 19$, cannot reject at 5%).

A spectral analysis of the cross-domain correlation matrix reveals structural clustering: arithmetic sequences form one connected component (Fiedler value $\lambda_2 = 0.273$) with Sato–Tate equidistribution as a weakly connected outlier.

Implemented in Rust (5,400 LOC conjecture engine, 1,300 LOC number theory module, 345+ tests, Z3 4.15.8 backend). Open source.

Keywords: automated conjecture, symbolic regression, formal verification, self-correcting AI, epistemic integrity, cross-domain pattern matching

1 Introduction

Automated conjecture generation has advanced significantly in recent years, from Eureqa's physical law discovery [Schmidt and Lipson, 2009] to DeepMind's knot theory insights [Davies et al., 2021] and the Ramanujan Machine's continued fraction identities [Raayoni et al., 2021]. However, existing systems share a critical weakness: *they cannot catch their own false positives.* A neural network that discovers a pattern has no mechanism to audit whether that pattern is a structural artifact, a model-fitting coincidence, or a genuine mathematical relationship.

We argue that trustworthy automated mathematics requires not just discovery but *epistemic self-correction*: the ability to detect and retract findings that are structurally trivial or statistically unsupported. We identify four requirements:

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1. **Grounded observation**: formulas discovered from computed data, not recalled from training corpora.

2. **Formal verification**: conjectures proven by an independent engine (Z3 SMT solver), not just numerically validated.

3. **Self-correction**: the system autonomously detects and retracts tautologies and model-dependent artifacts.

4. **Cross-domain pattern matching**: the system finds correspondences between different mathematical domains without being told what to look for.

The Ramanujan Protocol implements all four as a closed pipeline. We demonstrate it across number theory, combinatorics, dynamical systems, and Ramsey theory, with extended case studies in Langlands-type correspondence recovery and extreme value statistics.

2 Architecture

The pipeline consists of five stages:

Stage 1: Observe. Mathematical engines compute numerical sequences: $a_p(E)$ from elliptic curve point counting, $p(n)$ from partition enumeration, γ_k from Riemann zeta zero finding, etc. Each sequence carries domain metadata (NumberTheory, Physics, Combinatorics).

Stage 2: Discover. Grammar-guided genetic programming with ensemble search (3 seeds), template library seeding (growth-class analysis), and semantic crossover (rejects offspring $10\times$ worse than parents). Constants optimized by L-BFGS with finite-difference gradients. Pareto-optimal (MSE, complexity) frontier computed alongside scalar fitness.

Stage 3: Verify. Four verification levels: (i) numerical (held-out 20% test split), (ii) bounded induction ($n \leq 1000$), (iii) formal Z3 proof (SMTLIB2 QF_NIA/QF_NRA, quantifier-free nonlinear arithmetic), (iv) symbolic derivative verification ($dE/dt = 0$ for conservation laws).

Stage 4: Self-Correct. Competing model analysis (AIC comparison), bootstrap confidence intervals, structural tautology detection. Any finding that fails these audits is retracted.

Stage 5: Map. Cross-domain correlation matrix computed via Spearman rank correlation. Graph Laplacian eigenvalues extracted to reveal the spectral geometry of mathematical interconnection.

3 Results I: Formula Discovery and Proof

3.1 Closed-Loop Pipeline (Observe $\rightarrow$ Discover $\rightarrow$ Prove)

The complete pipeline was demonstrated on triangular numbers: the engine observed $\{1, 3, 6, 10, 15, \ldots\}$, discovered $T(n) = n(n+1)/2$ via GP, and Z3 proved the induction:

- **Base**: $T(1) = 1$ (Z3: UNSAT for negation).
- **Step**: $T(k+1) = T(k) + (k+1)$ (Z3: UNSAT for negation).

This constitutes a formal induction proof generated from raw numerical data, with no human guidance on what formula to search for or how to prove it.

Additional results: φ from Fibonacci ratios (MSE 8.4×10^{-21}), $E = x^2 + v^2$ conservation law (symbolic proof: $dE/dt = 2xv + 2v(-x) = 0$), Bell–Stirling identity ($B(n) = \sum S(n,k)$, exact to $n = 15$).

3.2 Formal Proofs via Z3

Statement	Method	Result
$T(n) = n(n+1)/2$	QF_NIA induction	**Proved**
$\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$	QF_NIA induction	**Proved**
$n^2 \geq n$ for $n \geq 1$	QF_NRA inequality	**Proved**
$(n + 1/n)/2 \geq 1$ (AM-GM)	QF_NRA inequality	**Proved**
$R(3,3) = 6$	SAT/UNSAT	**Proved**
$R(3,4) = 9$	SAT/UNSAT	**Proved**
$R(4,4) > 17$	SAT	**Verified**

4 Epistemic Self-Correction

The Protocol's most distinctive capability is not discovery but *retraction*. The system autonomously caught two false positives:

4.1 Structural Tautology: GCT Obstruction

The engine computed Kronecker coefficient obstruction ratios for same-weight partition triples and reported 99.9% obstruction at $n = 6$—apparently strong computational evidence. Post-hoc auditing revealed this is a **mathematical tautology**: the Littlewood–Richardson size constraint $|\mu| + |\nu| = |\lambda|$ forces $2w = w$ for same-weight triples, so $w = 0$ and all non-trivial triples vanish by construction. The finding was retracted before publication, preventing a false claim of evidence for $P \neq NP$.

4.2 Model Artifact: *abc* Quality Decay

The engine discovered $q(\text{rank}) \approx 1 + A/\text{rank}^B$ with exponent $B \approx 1.05$, stable across four scales ($c \leq$ 1K–50K). This appeared to be a universal power law. Competing model analysis revealed:

Model	R^2	AIC	Verdict
Exponential: $A \cdot e^{-B \cdot \text{rank}}$	**0.930**	-865	**Best**
Power law: A/rank^B	0.690	-270	Rejected
Log-corrected: $A/(n \ln^C n)$	0.000	-469	Rejected

The AIC difference of 596 is decisive. Bootstrap 95% CI for the power-law exponent: $B \in [0.92, 1.24]$—too wide to claim $B \approx 1.05$. The true decay is exponential, consistent with extreme value statistics.

4.3 Significance

These retractions are not failures—they are the system's primary contribution. Most automated discovery systems report findings and move on. The Ramanujan Protocol audits its own outputs via competing model analysis (AIC), bootstrap confidence intervals, and structural constraint checking. A system that cannot catch its own false positives cannot be trusted with its true positives.

5 Case Study: Cross-Domain Correspondence Recovery

As a test of the Protocol's cross-domain pattern matching capability, we applied it to the Langlands correspondence between elliptic curves and modular forms.

5.1 Modularity Correspondence Recovery

We generated $a_p(E) = p + 1 - \#E(\mathbb{F}_p)$ for 7 Cremona curves (conductors 11–21) and $c_n(f)$ for 7 weight-2 newforms via Hecke multiplicativity. The engine was given all 14 sequences plus 3 decoy forms with plausible but incorrect coefficients, *without pairing information.*

Result: 7/7 correct correspondences identified (95/95 prime matches), 0/3 decoy false positives. The engine recovered all known modularity correspondences from raw numerical data. This demonstrates that the Protocol can detect exact coefficient identities across mathematical domains without being told what to look for.

5.2 Additional Verifications

Sato–Tate distribution: Frobenius angles $\theta_p = \arccos(a_p/2\sqrt{p})$ for 302 primes on curve 11a1 confirm the $\sin^2\theta$ density shape.

Montgomery pair correlation: Zeta zero spacings tested against GUE: $\chi^2 = 9.28$ ($df = 19$, 5% critical ≈ 31.3). Cannot reject GUE at 5% significance. Level repulsion confirmed.

BSD critical values: $L(11\text{a}1, 1) = 0.2538$ (LMFDB: 0.2538, 4-digit match). All rank-0 curves satisfy $L(E, 1) > 0$.

6 Exploratory: Correlation Structure of Arithmetic Sequences

We computed Spearman rank correlations between six arithmetic sequences (prime gaps, *abc* quality, partitions, class numbers, zeta zero spacings, Sato–Tate angles; $n = 100$ each) and analyzed the resulting graph via spectral methods.

6.1 Correlation Structure

Sato–Tate angles are statistically independent of all other sequences ($|\rho| < 0.1$), confirming equidistribution and serving as a methodological control. The remaining five sequences form a correlated block with $|\rho| > 0.5$ for several pairs (e.g., class numbers $\leftrightarrow$ partitions: $\rho = 0.62$).

6.2 Laplacian Spectrum

The 6×6 correlation matrix (absolute values as edge weights) yields Graph Laplacian eigenvalues:

$$\lambda = \{0.000,\ 0.273,\ 1.860,\ 2.127,\ 2.716,\ 3.578\}$$

$\lambda_1 = 0$: all mathematics is **one connected component**. $\lambda_2 = 0.273$ (Fiedler value): the connection is **weak** (spectral gap $\lambda_2/\lambda_6 = 0.076$).

6.3 The Fiedler Clustering

The Fiedler vector (eigenvector of λ_2) partitions the sequences:

Sequence	Fiedler component	Cluster
Sato–Tate	+0.912	**A (isolated)**
Prime gaps	−0.195	B (arithmetic)
abc quality	−0.196	B
Partitions	−0.196	B
Class numbers	−0.161	B
Zeta spacings	−0.166	B

Interpretation: Arithmetic sequences (primes, partitions, class numbers, *abc* triples, zeta zeros) form one tightly coupled continent. Sato–Tate equidistribution is a separate island. The Langlands correspondence is the bridge between them—it maps the arithmetic structure of elliptic curves to the equidistributed angles of their Frobenius elements.

7 Related Work

Symbolic regression: Eureqa [Schmidt and Lipson, 2009], AI Feynman [Udrescu and Tegmark, 2020], PySR [Cranmer, 2023]. Our work adds formal Z3 verification and self-correction.

AI for mathematics: DeepMind's knot theory [Davies et al., 2021], AlphaProof [DeepMind, 2024], the Ramanujan Machine [Raayoni et al., 2021]. We achieve comparable cross-domain discovery with a transparent, deterministic pipeline (no neural networks).

Computational number theory: Cremona's tables [Cremona, 1997], LMFDB [The LMFDB Collaboration, 2024]. We verify modularity computationally and discover it autonomously.

8 Conclusion

The Ramanujan Protocol demonstrates that automated conjecture generation becomes trustworthy when paired with formal verification and epistemic self-correction. The system's principal contribution is not any specific formula it discovered, but the demonstration that an automated pipeline can:

- **Retract** its own false positives (GCT tautology, *abc* model artifact) before publication.
- **Recover** known cross-domain correspondences (modularity, 7/7 correct with 0 false positives) from raw data.
- **Verify** formally (Z3 induction proofs, Ramsey SAT) rather than just numerically.

The retractions are as important as the recoveries. A system that cannot catch its own false positives cannot be trusted with its true positives.

The pipeline is open-source (Rust, 345+ tests) and implemented as part of the Symthaea cognitive architecture.[1]

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